

6. Radish is a salad vegetable. The plants are grown in large quantities in greenhouses by commercial producers.

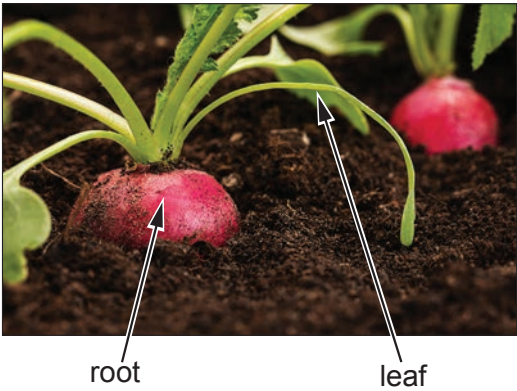
It is important that photosynthesis occurs at a high rate in the leaves of the plants to give the largest possible crop.

Photosynthesis requires carbon dioxide, water and light and produces glucose and oxygen.

Commercial greenhouse

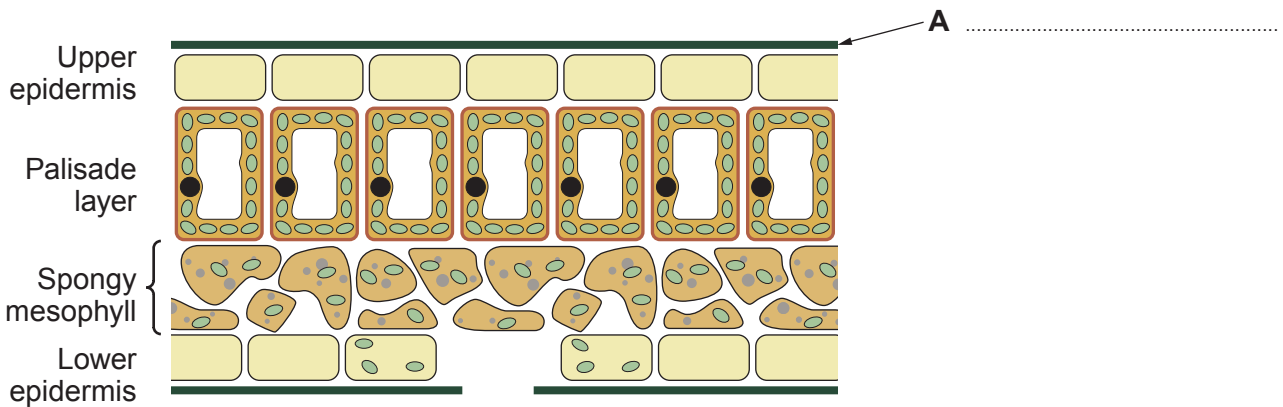


Radish plants



- (a) (i) The root of the radish contains stored starch. State the name of the molecules which are converted into starch. [1]

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The diagram below shows a section through a leaf of a plant such as radish.

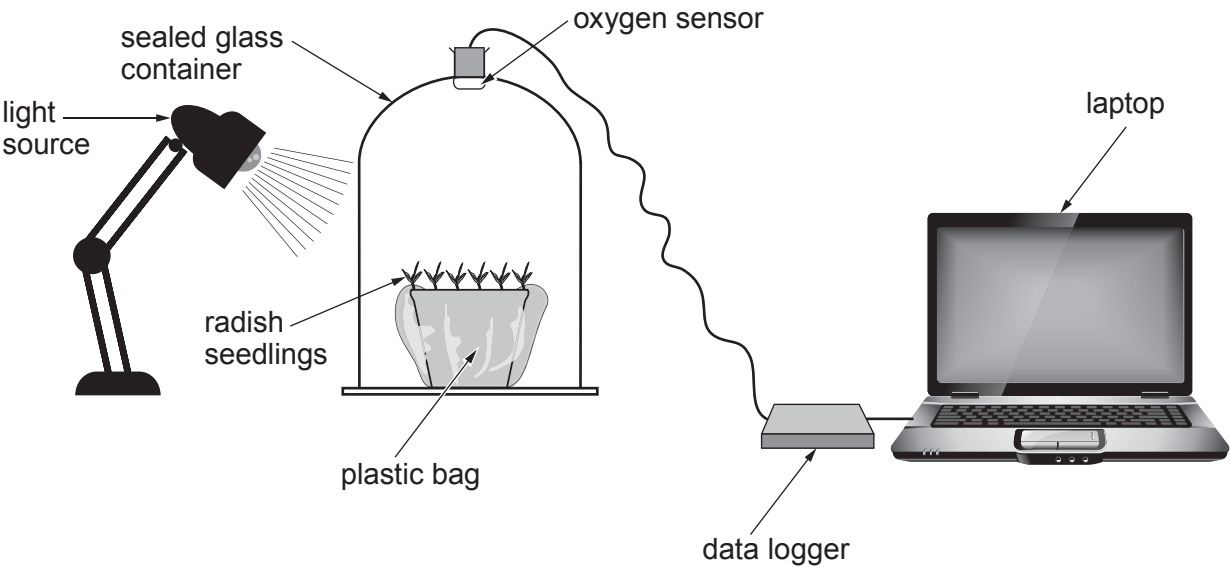


- (ii) I. Label structure A on the diagram. [1]
II. Explain why **most** photosynthesis occurs in the cells of the palisade layer. [3]

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(b) Robert, a commercial radish grower, investigated the effects of temperature on the growth of radish seedlings. He used a range of temperatures and used the apparatus shown below to measure oxygen production. From this he calculated the mean rate of photosynthesis at each temperature.



The results of his investigation are shown on the graph and in the table.

Temperature (°C)	Rate of photosynthesis (a.u.)
5	<i>Values shown on the graph</i>
10	
20	
25	51
30	60
35	64
40	59
45	24
55	15

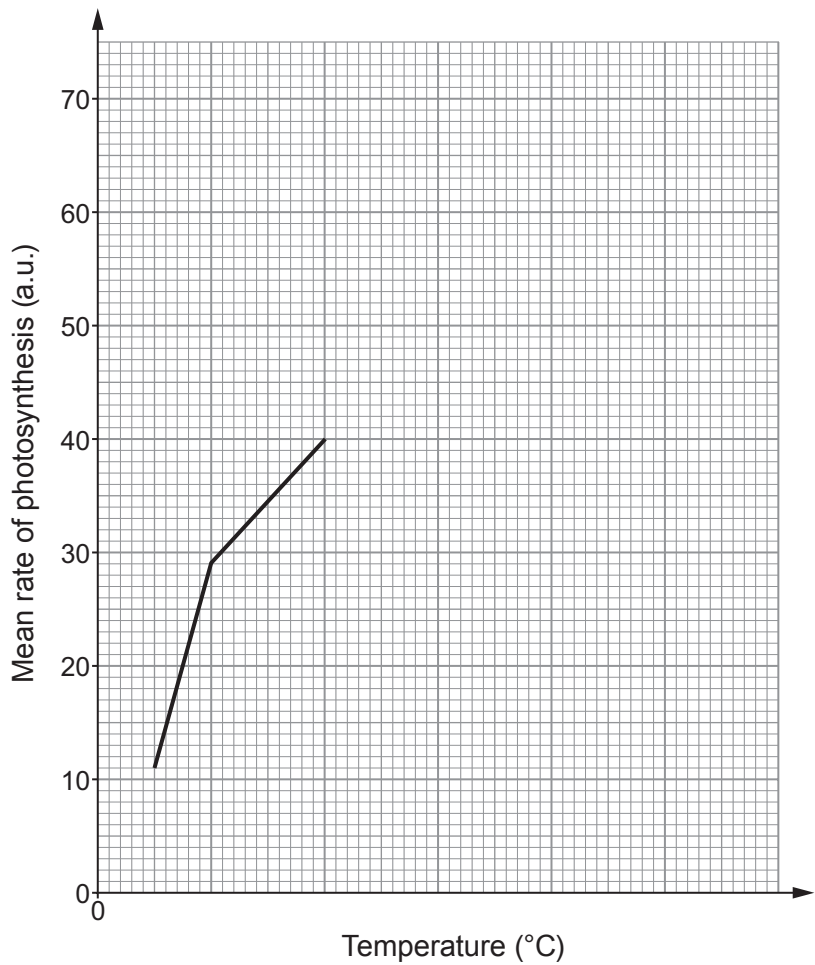
- (i) Complete the graph of results by: [4]
- I. adding the scale for temperature;

II. plotting the values from 25°C - 55°C;

III. joining the plots with a ruler to complete the line



Examiner
only



- (ii) From the graph,
- I. Describe how the increase in temperature affects the rate of photosynthesis. [2]
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 - II. Calculate the difference in the rate of photosynthesis between 5°C and 20°C. [1]

Difference = a.u.



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(iii) I. From these results state the optimum temperature for Robert to use for growing radish plants and explain your answer. [1]

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II. How could Robert improve his investigation so that the optimum temperature could be identified more accurately? [1]

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